Pocket Some MECHATRONIC TOOLS For The INDUSTRIAL IoT

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Electronic Workbench (design and simulation)

Electronic simulation software by nature, this tool is aimed at making you an expert in drawing electronic circuit schematics using an array of components super quickly. You can start by choosing the required components and start connecting these one by one to create circuits easily. The software lets you connect many measurement tools to your predesigned circuit and run the simulation even before your project is implemented in real time. The application runs quite well on Windows platform, and you can download this from this edition of *Electronics For You Plus* DVD.

HOTINT (design and simulation)

If you are looking for software that helps you simulate Physics based flexible, multi-body systems, then this software is for you. This mechatronics simulation software allows modelling of complex mechatronic and multi-body systems including dynamic and static computation, Eigen mode calculations, elements for rotor-dynamics, IoT blocks for building controllers, beams suitable of large deformations, parameter variation and optimisation.

HOTINT also supports coupling with controllers created with X2C, which is a free and open source tool for model based development and code generation in real-time control algorithms using microprocessor units. X2C uses open source environment of Scilab/Xcos for building the graphical control model. This software is only available for Windows platform.

Some popular resources

DisplayCAL. DisplayCAL, previously called DispcalGUI, is a graphical user interface developed for the purpose of display calibration and profiling using Argyll CMS, which is an open source colour-management system. Using DisplayCAL, you can calibrate and characterise display devices using hardware sensors and play around with available settings like customisable white point, luminance and tone response curve as well as create matrix and look-up-table ICC profiles, with optional gamut mapping, as well as some proprietary 3D LUT formats. DisplayCAL is written in Python, and has been packaged for all supporting operating system platforms.

Apache OpenOffice. Apache OpenOffice is an open source office productivity software suite. It contains a word processor (Writer), a spreadsheet (Calc), a presentation application (Impress), a drawing application (Draw), a formula editor (Math) and a database management application (Base).

UNetbootin. UNetbootin allows you to create bootable live USB drives for Ubuntu, Fedora and other Linux distributions without burning it onto a DVD or CD. This software also lets you select and download many distributions of operating systems in its out-of-the-box repository.

PCB Creator (PCB)

PCB Creator is your one-stop solution for PCB layout and schematic capturing. With the capability to create two- to four-layer custom PCBs, new parts and footprints could be designed within a few hours of getting your hands on this software. You can import and export your designs and libraries with other EDA tools, use Schematic Capture and preview the designs. The software is powered by DipTrace platform and provides a four-in-one design environment that lets you customise designs and create PCB layouts with auto router, 3D PCB preview, and component and pattern editors. It also supports real-time 3D PCB preview and export.

TANGO control system (SCADA)

TANGO is an open source toolkit for building high-performance and high-quality distributed control systems for small and large installations. The toolkit design is based on the concept of distributed devices or objects, and provides native support for multiple programming languages. It implements a full set of tools for developing, managing and monitoring small and large control systems.

KnightOS (utility)

Some of us like using operating systems to their full potential. To do that there must be an operating system that matches the level of freedom, customisation and agility required. KnightOS is designed to run calculators or emulators on Z80 CPUs. This platform also gives you the freedom to use Assembly, C, Python, HTML/CSS and JavaScript, and use custom kernels during development. EFY

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